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The required derivative is $dx/dt = 0 + A_1 + 2A_2t$

The second derivative is $d^2x/d^2t = 2A_2$

## 2.9 INTEGRAL CALCULUS

Integration is basically an area finding process. For certain geometric shapes we can directly find the area. But for irregular shapes the process of integration is used. Consider for example the areas of a rectangle and an irregularly shaped curve, as shown in Figure 2.29.

The area of the rectangle is simply given by A = length × breadth = (b–a) c

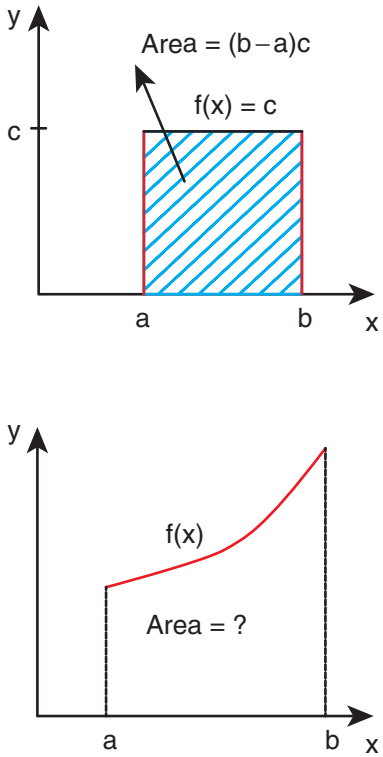



**Figure 2.29** Area of rectangular and irregular shape

But to find the area of the irregular shaped curve given by f(x), we divide the area into rectangular strips as shown in the Figure 2.30.

The area under the curve is approximately equal to sum of areas of each rectangular strip.

This is given by $A \approx f(a)\Delta x + f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x$.

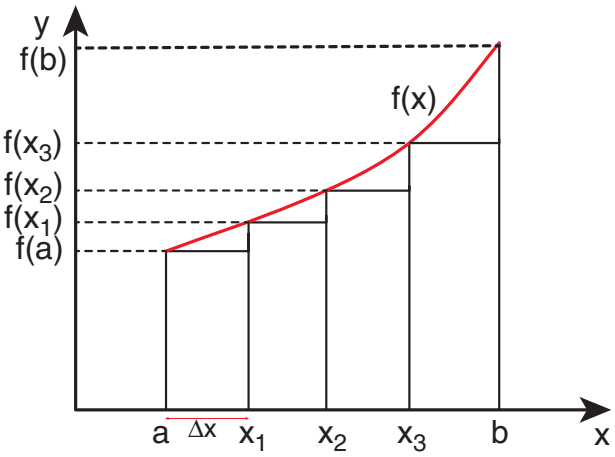



**Figure 2.30** Area under the curve using rectangular strip

Where $f(a)$ is the value of the function $f(x)$ at x = a, $f(x_1)$ is the value of $f(x)$ for $x = x_1$ and so on.

As we increase the number of strips, the area evaluated becomes more accurate. If the area under the curve is divided into N strips, the area under the curve is given by

$$A = \sum_{n=1}^{N} f(x_n)\Delta x_n$$

As the number of strips goes to infinity, $N \to \infty$, the sum becomes an integral,

$$A = \int_a^b f(x)\,dx$$

(Note: As $N \to \infty$, $\Delta x \to 0$)

The integration will give the total area under the curve f (x). This is shown in Figure 2.31.